



NAGRAJ METAL INDUSTRIES

ISO CERTIFIED COMPANY

STOCKISTS • SUPPLIER • DEALERS

WHY CHOOSE US?

- ✓ Wide Range of Quality Products
- ✓ Timely Delivery
- ✓ Competitive Prices
- ✓ Trusted by Industries Across India
- ✓ Strong Network & Reliability
- ✓ Customer Satisfaction is Our Priority

*"Your Trusted Partner in **Quality, Reliability & Service.**"*

DEALING IN ALL TYPES OF FERROUS & NON FERROUS METALS



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OUR ADDRESS

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Mumbai 400004



TRUST



QUALITY



RELIABILITY



COMMITMENT

• THANK YOU FOR VISITING US •



STAINLESS STEEL

NAGRAJ METAL INDUSTRIES



PREMIUM QUALITY STAINLESS STEEL PRODUCTS

1. STAINLESS STEEL SHEET



- ✓ Smooth Surface Finish
- ✓ High Corrosion Resistance
- ✓ Excellent Formability
- ✓ High Strength & Durability
- ✓ Available in Various Thicknesses

2. STAINLESS STEEL PLATE



- ✓ Thick & Heavy Duty Plates
- ✓ Superior Strength
- ✓ Corrosion & Heat Resistant
- ✓ Precise Dimensions
- ✓ Ideal for Structural & Industrial Use

3. STAINLESS STEEL COIL



- ✓ High Quality Stainless Steel
- ✓ Excellent Corrosion Resistance
- ✓ Uniform Thickness & Finish
- ✓ High Tensile Strength
- ✓ Available in Multiple Grades & Widths

4. STAINLESS STEEL SLITTING



- ✓ Precision Slitting for Exact Width
- ✓ Smooth Edges & Surface
- ✓ Custom Widths Available
- ✓ High Efficiency Processing
- ✓ Ideal for Industrial Applications



PREMIUM
QUALITY



SS 304 / SS 316
AVAILABLE



HIGH STRENGTH
& DURABLE



CUSTOM SIZE &
THICKNESS



WIDE RANGE OF
APPLICATIONS



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STAINLESS STEEL

• NAGRAJ METAL INDUSTRIES •



PREMIUM QUALITY STAINLESS STEEL PRODUCTS

RING



- ✓ High Strength
- ✓ Corrosion Resistant
- ✓ Long Service Life
- ✓ Accurate Dimensions
- ✓ Smooth Surface Finish

CIRCLE



- ✓ Solid & Durable
- ✓ Excellent Corrosion Resistance
- ✓ High Strength & Toughness
- ✓ Perfect Roundness
- ✓ Smooth & Shiny Finish

SQUARE FRAME



- ✓ Strong & Sturdy
- ✓ Rust & Corrosion Resistant
- ✓ High Dimensional Accuracy
- ✓ Neat Welding / Excellent Finish
- ✓ Ideal for Industrial & Decorative Use



SS 304 /
SS 316 Available



Food Grade
Quality



Superior
Surface Finish



Custom Size
Available



Wide Range of
Industrial Applications



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TYPES OF PIPES

NAGRAJ METAL INDUSTRIES



PREMIUM QUALITY METAL PIPES – STRENGTH, DURABILITY & TRUST

ROUND PIPE



- ✓ Stainless Steel Round Pipe
- ✓ ERW Round Pipe
- ✓ Seamless Round Pipe
- ✓ Polished Round Pipe
- ✓ Structural Round Pipe
- ✓ Hydraulic Round Pipe

COPPER PIPE



- ✓ Seamless Copper Pipe
- ✓ Straight Copper Pipe
- ✓ Pan Cake Copper Pipe
- ✓ Copper Round Tube
- ✓ Hard Drawn Copper Pipe
- ✓ Soft Copper Pipe

BRASS PIPE



- ✓ Seamless Brass Pipe
- ✓ Extruded Brass Pipe
- ✓ Cut Brass Pipe
- ✓ Brass Round Tube
- ✓ Brass Hollow Pipe
- ✓ Brass Instrument Pipe

ALUMINIUM PIPE



- ✓ Aluminium Round Pipe
- ✓ Aluminium Square Pipe
- ✓ Aluminium Rectangular Pipe
- ✓ Aluminium Hollow Pipe
- ✓ Aluminium Welded Pipe
- ✓ Aluminium Seamless Pipe

OTHER TYPES OF PIPES

SQUARE PIPE



RECTANGULAR PIPE



GI PIPE



BLACK PIPE



WELDED PIPE



SEAMLESS PIPE



SCH 40 / SCH 80 PIPE



THREADED PIPE



ALL TYPES OF PIPES – ONE TRUSTED SOURCE



PREMIUM
QUALITY



CORROSION
RESISTANT



HIGH
STRENGTH



ACCURATE
DIMENSIONS



WIDE RANGE OF
APPLICATIONS



READY STOCK &
CUSTOM SIZE



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TYPES OF ROUND BAR



NAGRAJ METAL INDUSTRIES

1. EN9 ROUND BAR



2. EN8 ROUND BAR



3. MS ROUND BAR



4. SS 304 ROUND BAR



5. SS 316 ROUND BAR



6. ALUMINIUM ROUND BAR



7. BRASS ROUND BAR



8. COPPER ROUND BAR



9. TITANIUM ROUND BAR



10. ALLOY STEEL ROUND BAR



11. TOOL STEEL ROUND BAR



12. CARBON STEEL ROUND BAR



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TYPES OF FLANGES



NAGRAJ METAL INDUSTRIES

1. WELD NECK FLANGE



2. SLIP-ON FLANGE



3. BLIND FLANGE



4. LAP JOINT FLANGE



5. SOCKET WELD FLANGE



6. THREADED FLANGE



7. REDUCING FLANGE



8. EXPANDING FLANGE



9. ORIFICE FLANGE



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TYPES OF COPPER FITTINGS



NAGRAJ METAL INDUSTRIES

HIGH QUALITY COPPER | CORROSION RESISTANT | LEAK PROOF | DURABLE & LONG LASTING

PREMIUM QUALITY COPPER FITTINGS FOR ALL YOUR PIPING NEEDS

1 COPPER EQUAL COUPLING  EQUAL COUPLING	2 COPPER REDUCING COUPLING  REDUCING COUPLING	3 COPPER 90° ELBOW  90° ELBOW	4 COPPER 45° ELBOW  45° ELBOW	5 COPPER STREET ELBOW  STREET ELBOW
6 COPPER EQUAL TEE  EQUAL TEE	7 COPPER REDUCING TEE  REDUCING TEE	8 COPPER EQUAL CROSS  EQUAL CROSS	9 COPPER END CAP  END CAP	10 COPPER END FEED CAP  END FEED CAP
11 COPPER 90° RETURN BEND  90° RETURN BEND	12 COPPER U BEND  U BEND	13 COPPER COUPLING WITH STOP  COUPLING WITH STOP	14 COPPER MALE ADAPTER  MALE ADAPTER	15 COPPER FEMALE ADAPTER  FEMALE ADAPTER
16 COPPER UNION  UNION	17 COPPER REDUCING UNION  REDUCING UNION	18 COPPER FLANGE ADAPTER  FLANGE ADAPTER	19 COPPER SOLDER RING  SOLDER RING	20 COPPER PIPE NIPPLE  PIPE NIPPLE
21 COPPER HEX NIPPLE  HEX NIPPLE	22 COPPER PLUG (HEX)  PLUG (HEX)	23 COPPER BUSH (REDUCING)  BUSH (REDUCING)	24 COPPER FERRULE  FERRULE	25 COPPER SADDLE  SADDLE



BEST QUALITY



STRONG & RELIABLE



MADE TO LAST

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ALL TYPES OF INDUSTRIAL RAW MATERIALS | FERROUS & NON FERROUS METALS



TYPES OF BRASS FITTINGS

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PREMIUM QUALITY BRASS FITTINGS – STRENGTH, DURABILITY & CORROSION RESISTANCE

ELBOW 90°



ELBOW 45°



TEE



REDUCING TEE



COUPLING



REDUCING COUPLING



UNION



MALE ADAPTER



FEMALE ADAPTER



HEX NIPPLE



BUSH



PLUG



CAP



CROSS



BARB HOSE FITTING



COMPRESSION FITTING



FLARE FITTING



SLEEVE



BULKHEAD FITTING



GARDEN HOSE ADAPTER



Y-STRAINER



VALVE (BALL VALVE)



VALVE (GATE VALVE)



VALVE (CHECK VALVE)



PREMIUM QUALITY
BRASS



CORROSION
RESISTANT



HIGH
STRENGTH



ACCURATE
THREADS



EXCELLENT
FINISH



WIDE RANGE OF
APPLICATIONS



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TYPES OF BRASS FITTINGS

NAGRAJ METAL INDUSTRIES



PT SOCKET



SOCKET



VALVE SOCKET



VALVE ELBOW



PT ELBOW



ELBOW



PT TEE



ELBOW LUGAGE



TEE



NOTE: Brass Fittings are used in Plumbing, Gas, Water, Oil & Industrial Applications.



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PREMIUM
QUALITY



HIGH GRADE
BRASS



RUST & CORROSION
RESISTANT



STRONG &
DURABLE



LONG LASTING
PERFORMANCE



TYPES OF NUTS



NAGRAJ METAL INDUSTRIES

1 HEX NUT



- Most common type. Used with bolts and screws for general applications.

2 NYLOCK NUT (NYLON INSERT)



- Has a nylon ring to prevent loosening due to vibration.

3 WING NUT



- Easy to tighten and loosen by hand. Used in frequent assembly/disassembly.

4 CAP NUT (ACORN NUT)



- Closed end nut used to cover the bolt end and provide a neat appearance.

5 FLANGE NUT



- Built-in washer flange distributes load and reduces loosening.

6 LOCK NUT (ALL METAL)



- All-metal design resists loosening under vibration and torque.

7 SQUARE NUT



- Four sides provide better grip. Used in heavy-duty applications.

8 CASTLE NUT (SLOTTED NUT)



- Used with a cotter pin to lock the nut in place. Common in axles.

9 T NUT



- Pronged nut used in wood or soft materials to provide a strong threaded anchor.

10 HEX JAM NUT



- Thin nut used to lock the position of other nuts or adjust clearances.

11 COUPLING NUT



- Long nut used to join two threaded rods or extend the length of a bolt.

12 LOCK NUT (SERRATED)



- Serrated teeth grip the surface to prevent loosening due to vibration.



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**TRUSTED
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TYPES OF LOCK NUTS



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1 NYLON INSERT LOCK NUT



Nylon insert prevents loosening due to vibration.

2 ALL METAL LOCK NUT



All-metal design provides high strength locking.

3 JAM NUT



Used with other nuts to prevent loosening.

4 SERRATED FLANGE NUT



Serrated flange provides better grip and locking.

5 FLANGE LOCK NUT



Flange distributes load and prevents loosening.

6 CASTLE NUT (CROWN NUT)



Used with a cotter pin for secure locking.

7 SLOTTED NUT



Slots allow use of a cotter pin.

8 CAP NUT (ACORN NUT)



Protects bolt end and provides a decorative finish.

9 WING NUT



Easy hand tightening and removal.

10 KEP NUT



External teeth bite into surface for locking.

11 TOOTHED LOCK NUT



Teeth grip the mating surface to lock.

12 SPRING LOCK NUT



Spring action maintains constant tension.

13 TOP LOCK NUT



Top locking feature prevents loosening.

14 PREVAILING TORQUE LOCK NUT



Prevailing torque resists loosening under vibration.

15 CONE LOCK NUT



Conical seat ensures tight and secure fit.

16 CENTER LOCK NUT



Center locking for better hold.

17 PAL NUT



Used with a lock washer for secure fastening.

18 SELF-LOCKING NUT (DEFORMED THREAD)



Deformed threads create strong locking action.

19 VIBRATION PROOF LOCK NUT



Special design resists vibration and loosening.

20 HEX COUPLING LOCK NUT



Used to connect threaded rods and prevent loosening.



NOTE: Lock nuts are used in various applications to prevent loosening caused by vibration and torque.



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HIGH STRENGTH
& DURABILITY



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TYPES OF BOLTS



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1 HEX BOLT



2 CARRIAGE BOLT



3 SOCKET HEAD CAP BOLT



4 COUNTERSUNK BOLT



5 FLANGE BOLT



6 EYE BOLT



7 U BOLT



8 J BOLT



9 LAG BOLT



10 T BOLT



11 WHEEL BOLT



12 ANCHOR BOLT



NOTE: Bolts are used in various applications for strong, secure and reliable fastening.



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TYPES OF EYE BOLTS



NAGRAJ METAL INDUSTRIES

1. SHOULDER EYE BOLT



Has an unthreaded shoulder under the eye.

2. PLAIN EYE BOLT



Standard eye bolt without shoulder.

3. COLLARED EYE BOLT



Has a collar (flange) under the eye.

4. SHOULDERLESS EYE BOLT



No shoulder or collar, thread runs up to the eye.

5. LIFTING EYE BOLT



Designed for lifting heavy loads safely.

6. SWIVEL EYE BOLT



Eye rotates 360°, allows full load alignment.

7. HOIST RING BOLT



Heavy-duty ring for hoisting and rigging.

8. FEMALE EYE BOLT



Has an internal (female) threaded hole.

9. WELD ON EYE BOLT



Base is welded to the structure.

10. LONG SHANK EYE BOLT



Extended threaded length for deep fastening.

11. JIS EYE BOLT (JAPANESE TYPE)



Manufactured as per Japanese Industrial Standards (JIS).

12. DIN EYE BOLT (GERMAN TYPE)



Manufactured as per German Standards (DIN).

13. UNC EYE BOLT (COARSE THREAD)



Unified National Coarse thread (UNC).

14. UNF EYE BOLT (FINE THREAD)



Unified National Fine thread (UNF).

15. STAINLESS STEEL EYE BOLT



Corrosion-resistant, made from stainless steel.

Note: Eye bolts are used for lifting, rigging, securing, and various industrial applications.



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TYPES OF WASHERS

NAGRAJ METAL INDUSTRIES



1. FLAT WASHER



2. SPRING (SPLIT) WASHER



3. LOCK WASHER



4. TOOTH LOCK WASHER



5. INTERNAL TOOTH WASHER



6. STAR WASHER



7. CUP WASHER



8. FINISHING WASHER



9. BELLEVILLE (DISC SPRING) WASHER



10. WAVE WASHER



11. FENDER WASHER



12. SHOULDER WASHER



13. SEALING WASHER



14. INSULATING WASHER



15. THRUST WASHER



16. SQUARE WASHER



 **NOTE:** Washers are used to distribute load, reduce friction, prevent loosening and protect surfaces.



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TYPES OF SCREWS



NAGRAJ METAL INDUSTRIES

— STRONG. RELIABLE. PRECISION ENGINEERED. —

1 MACHINE SCREW



Used with a nut or in a threaded hole.

2 SELF-TAPPING SCREW



Creates its own threads in the material.

3 WOOD SCREW



Designed for use in wood and soft materials.

4 SHEET METAL SCREW



Used to fasten thin metal sheets together.

5 HEX SCREW



High strength screw with hexagonal head.

6 CARRIAGE BOLT



Used with a nut, prevents rotation when tightened.

7 SOCKET HEAD CAP SCREW



High strength with hex socket drive.

8 COUNTERSUNK SCREW



Flat head for a flush finish.

9 PAN HEAD SCREW



Rounded head with a slightly elevated top.

10 ROUND HEAD SCREW



Domed head for a smooth, finished look.

11 FLAT HEAD SCREW



Flat head sits flush with the surface.

12 OVAL HEAD SCREW



Low-profile head for a neat appearance.

13 BUTTON HEAD SCREW



Low, rounded head with a wide bearing surface.

14 SET SCREW



Used to secure one object inside another.

15 GRUB SCREW



Short set screw used without a head.

16 THUMB SCREW



Hand-tightened screw for frequent adjustment.

17 WING SCREW



Wing-shaped head for easy hand tightening.

18 EYE SCREW



Used for attaching cables, ropes or chains.

19 LAG SCREW



Heavy-duty screw for wood and lagging.

20 CONCRETE SCREW



Designed for fastening into concrete or masonry.

21 ANCHOR SCREW



Expands in concrete for secure anchoring.

22 DRYWALL SCREW



Used to fasten drywall to wood or metal studs.

23 SECURITY SCREW



Special head design for tamper resistance.

24 SEMS SCREW



Screw with attached washer for added grip.

 **NOTE:** Screws are used in various applications for strong, secure and reliable fastening.



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HIGH STRENGTH & DURABILITY



CORROSION RESISTANT



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TYPES OF COTTER PINS



NAGRAJ METAL INDUSTRIES

USED FOR SECURING NUTS, CASTLE NUTS AND COMPONENTS

1 SPLIT COTTER PIN



Most common type. Used to secure nuts and components by inserting through a drilled hole.

2 HITCH COTTER PIN



Used for quick locking and removal in agricultural and industrial applications.

3 HAIR PIN COTTER PIN



Used in hydraulic cylinders, brake systems and automotive applications.

4 CLEVIS COTTER PIN



Used with clevis pins to prevent axial movement in mechanical assemblies.

5 BOW COTTER PIN



Stronger than standard pins. Ideal for heavy duty and high vibration areas.

6 SAFETY COTTER PIN



Designed for permanent locking applications. Provides extra safety and reliability.

7 ROUND COTTER PIN



Made from round wire. Used in general fastening and light duty applications.

8 SQUARE COTTER PIN



Square body provides better grip and prevents rotation. Used in heavy machinery.

9 DOUBLE LOOP COTTER PIN



Provides extra holding strength. Suitable for critical and vibration prone applications.

10 R-CLIP COTTER PIN



Also known as R-pin. Used for quick locking and securing shafts and components.

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TYPES OF BW PIPE FITTINGS



NAGRAJ METAL INDUSTRIES

— STRONG. RELIABLE. PRECISION ENGINEERED. —

1 90° ELBOW



2 45° ELBOW



3 180° RETURN BEND



4 TEE (EQUAL)



5 TEE (REDUCING)



6 CONCENTRIC REDUCER



7 ECCENTRIC REDUCER



8 CAP (END CAP)



9 CROSS



10 LATERAL



11 STUB END



12 WELD OLET



13 SOCKOLET



14 THREDOLET



15 NIPOLET



16 SWAGE NIPPLE



17 BOSS



18 INSERT (BOSS INSERT)



19 BULGED TEE



20 BULGED ELBOW



NOTE: BW (Butt Weld) pipe fittings are used to change direction, branch, reduce size or cap the piping system.



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TYPES OF PIPE FITTINGS



NAGRAJ METAL INDUSTRIES

1 ELBOW



2 90° STREET ELBOW



3 45° ELBOW



4 TEE



5 REDUCING TEE



6 COUPLING



7 REDUCING COUPLING



8 REDUCING NIPPLE



9 UNION



10 HEX NIPPLE



11 CAP



12 PIPE PLUG



13 BUSHING



14 COUNTER BORE



15 LATERAL



16 Y-STRAINER



17 SWAGE NIPPLE



18 FLANGE



19 REDUCING COUPLING



20 LOCK NUT





TYPES OF SS SOCKETS



NAGRAJ METAL INDUSTRIES

1 SS SOCKET



2 SS REDUCING SOCKET



3 SS HALF SOCKET



4 SS LONG SOCKET



5 SS STREET SOCKET



6 SS REDUCING STREET SOCKET



7 SS COUPLING SOCKET



8 SS UNION SOCKET



9 SS ELBOW SOCKET (90°)



10 SS REDUCING ELBOW SOCKET (90°)



11 SS TEE SOCKET



12 SS REDUCING TEE SOCKET



13 SS CROSS SOCKET



14 SS 45° ELBOW SOCKET



15 SS CAP SOCKET



16 SS HEX SOCKET (BUSH)



17 SS PLUG SOCKET



18 SS REDUCING BUSH SOCKET



19 SS FLANGE SOCKET



20 SS WELD SOCKET



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& DURABILITY



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STAINLESS STEEL SOCKETS FOR STRONG, SECURE & LEAK-PROOF CONNECTIONS



TYPES OF TC FITTINGS

NAGRAJ METAL INDUSTRIES



HYGIENIC • CORROSION RESISTANT • EASY TO ASSEMBLE • LEAK PROOF

1 90° ELBOW



Changes direction of flow by 90°.

2 45° ELBOW



Changes direction of flow by 45°.

3 180° U-BEND



Changes direction of flow by 180°.

4 TEE



Used to split or combine flow in two directions.

5 CROSS



Used to split or combine flow in four directions.

6 REDUCING TEE



Connects pipes of different sizes in a T connection.

7 Y TYPE TEE



Used to split or combine flow at an angle.

8 CONCENTRIC REDUCER



Connects pipes of different sizes in a straight line.

9 ECCENTRIC REDUCER



Connects pipes of different sizes with level offset.

10 FERRULE



Welded to the pipe end for TC clamp connection.

11 CAP



Used to close the end of a pipeline.

12 VACUUM BREAKER



Prevents back siphoning in hygienic systems.

13 SAMPLE VALVE



Used to take samples from the pipeline.

14 SPHERICAL STRAINER



Removes solid particles from the flow.

15 SIGHT GLASS



Allows visual inspection of the flow.

16 PIPE



Used to connect fittings and for straight flow.

17 Y TYPE STRAINER



Filters out impurities from the flow.

18 CLAMP



Used to connect and secure TC fittings.

19 GASKET / SEAL



Ensures leak proof sealing between joints.

20 TC HOSE ADAPTER



Connects hose to TC fittings.



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TYPES OF TC FITTINGS

NAGRAJ METAL INDUSTRIES



HYGIENIC • CORROSION RESISTANT • EASY TO ASSEMBLE • LEAK PROOF

1 IMPORTED FERRULE



Used to connect TC fittings with clamps.

2 FERRULE MALE / FEMALE



Male and female ferrules for TC connections.

3 FERRULE BLIND



Used to close the end of a TC connection.

4 SMS LINER / PART



Liner or part used in SMS fittings.

5 WELDABLE UNION



Used for easy union and disconnection.

6 CONICAL DISK FILTER



Filters particles in pipelines.

7 TC CLAMP



Clamps to secure TC fittings and joints.

8 HIGH PRESSURE CLAMP



Used in high pressure TC applications.

9 TRICLOVER CLAMP (UNION) INDIAN SET



Indian type tri-clamp union set.

10 TC HOSE



Connects hose to TC fittings.

11 DAIRY PIPE HOLDER



Holds dairy pipes securely.

12 IMPORTED PIPE HOLDER CLAMP



Imported clamp for pipe holding.

13 RUBBER GASKET



Ensures leak proof seal between joints.

14 BSP STAND



Support stand for BSP fittings.

15 BSP FERRULE



Ferrule with BSP thread connection.

16 TC TEE



Tee for TC pipe connections.

17 TC REDUCER



Reduces pipe size in TC system.

18 90° TC BEND



Changes direction of flow by 90°.

19 90° PLAIN BEND



90° bend for smooth flow direction.

20 TEE



Used to split or combine flow in pipelines.

21 REDUCER



Reduces pipe size in straight line.

22 Y TYPE TEE



Y shape tee for branch connection.

23 SMS UNION



Union for easy connection.

24 SMS BLIND NUT



Nut used to close SMS connection.

25 SIGHT GLASS



Allows visual inspection of the flow.



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TYPES OF HOSE CLAMPS



—●— NAGRAJ METAL INDUSTRIES —●—

1 WORM DRIVE HOSE CLAMP



- Most common type
- Easy to install & remove
- Used in general applications

2 T-BOLT HOSE CLAMP



- High clamping strength
- For heavy duty applications
- Used in automotive, turbo and industrial hoses

3 SPRING LOADED HOSE CLAMP



- Maintains constant pressure
- Auto-adjusts with temperature changes
- Used in fuel, coolant & vacuum lines

4 EAR (OETIKER) HOSE CLAMP



- Stepless design
- Compact & tamper-proof
- Used in constant or high vibration areas

5 HEAVY DUTY WORM DRIVE CLAMP



- Wider band for more strength
- Better sealing and durability
- Used in high pressure applications

6 CONSTANT TENSION HOSE CLAMP



- Provides constant tension
- Reduces risk of over-tightening
- Ideal for rubber or soft hoses

7 BAND-IT HOSE CLAMP



- Stainless steel band
- High clamping force
- Used in harsh environments and heavy duty applications

8 QUICK RELEASE HOSE CLAMP



- Tool-free quick release
- Reusable and convenient
- Used where frequent assembly is required

9 WIRE HOSE CLAMP



- Made from spring wire
- Light weight and economical
- Used in low pressure applications

10 BOLT CLAMP



- Strong bolt & nut tightening
- High strength and reliable
- Used in heavy duty and industrial hoses

11 PINCH BOLT HOSE CLAMP



- Compact and sturdy design
- Good sealing with uniform compression
- Used in automotive systems

12 MULTI-BAND HOSE CLAMP



- Multiple bands for even pressure
- Helps prevent leaks
- Used in soft or thick-walled hoses

13 STEPLESS HOSE CLAMP



- Stepless adjustment
- 360° uniform sealing
- Vibration resistant

14 PLASTIC COATED HOSE CLAMP



- Protects hose from damage
- Corrosion resistant
- Used in plumbing and automotive applications

15 CLIP-ON HOSE CLAMP



- Easy snap-on installation
- No tools required
- Used in air, fuel and low pressure lines

16 DOUBLE WIRE HOSE CLAMP



- Double wire for extra strength
- High torque tightening
- Used in high pressure and heavy duty hoses



TYPES OF INDUSTRIAL HOSE



NAGRAJ METAL INDUSTRIES

1. AIR HOSE



Used for conveying compressed air in pneumatic tools, compressors and industrial applications.

2. WATER HOSE



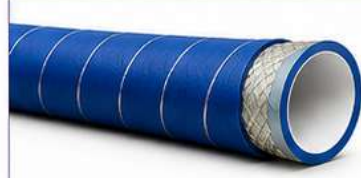
Used for general water transfer, irrigation, cooling and wash down applications.

3. FUEL HOSE



Designed for transfer of fuels like petrol, diesel, kerosene and other petroleum products.

4. CHEMICAL HOSE



Used for transferring chemicals, acids, alkalis and other corrosive substances.

5. STEAM HOSE



Built to handle high temperature saturated steam in industrial and process applications.

6. OIL HOSE



Used for suction and delivery of lubricating oils, hydraulic oils and other oil-based products.

7. HYDRAULIC HOSE



Used in hydraulic systems to transmit pressurized hydraulic fluid with high reliability.

8. SUCTION HOSE



Designed for suction and discharge of water, slurries, sand and other liquid materials.

9. FOOD GRADE HOSE



Used for transfer of food, beverages, dairy products and potable water in a hygienic manner.

10. ABRASION RESISTANT HOSE



Built to withstand wear and tear from abrasive materials like sand, gravel, cement and powders.

11. PTFE HOSE



Chemically inert and high temperature resistant hose for aggressive chemicals and critical applications.

12. EXHAUST HOSE



Used to remove engine exhaust fumes, smoke and gases in automotive and industrial systems.

13. RUBBER HOSE



General purpose hose used for air, water and mild chemical applications.

14. PVC HOSE



Lightweight and flexible hose used for water, air and chemical transfer in general applications.

15. FIRE HOSE



Designed for high pressure water discharge in fire fighting and emergency applications.

16. DUCT HOSE



Used for ventilation, fume extraction, dust collection and air circulation applications.

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TYPES OF VALVES



NAGRAJ METAL INDUSTRIES

STRONG. RELIABLE. PRECISION ENGINEERED.

1 GATE VALVE



Used to start or stop the flow. Best for on/off service with minimal pressure drop.

2 GLOBE VALVE



Used to regulate flow. Provides throttling action and good shut-off.

3 BALL VALVE



Uses a ball with a hole to control flow. Quarter-turn operation for quick on/off.

4 BUTTERFLY VALVE



Uses a rotating disc to control flow. Compact, lightweight and economical.

5 CHECK VALVE



Allows flow in one direction only. Automatically prevents reverse flow.

6 NEEDLE VALVE



7 PLUG VALVE



Uses a cylindrical or tapered plug with a hole to control flow.

8 DIAPHRAGM VALVE



Uses a flexible diaphragm to control flow. Ideal for corrosive or dirty fluids.

9 SAFETY VALVE



Automatically releases pressure when it exceeds set limit to prevent system damage.

10 PRESSURE RELIEF VALVE



Releases excess pressure from a system and returns to normal when pressure drops.

11 SOLENOID VALVE



Electrically operated valve used to control the flow of fluids or gases.

12 CONTROL VALVE



Used to regulate flow, pressure, temperature or level automatically in a system.

13 PINCH VALVE



Uses a flexible sleeve to control flow. Ideal for abrasive and corrosive media.

14 ANGLE VALVE



Flow changes direction by 90 degrees. Used in compact piping systems.

15 PISTON VALVE



Uses piston movement to control flow. Suitable for high-pressure applications.

KEY FEATURES

- Regulate or control flow effectively
- Ensure safety and system protection
- Improve efficiency and performance
- Reliable and durable construction
- Available in various materials for different applications

COMMON APPLICATIONS

- Oil & Gas
- Chemical Plants
- Treatment
- Pharmaceutical
- Power Plants
- HVAC Systems

COMMON ACTUATORS USED

- Manual Actuator
- Electric Actuator
- Pneumatic Actuator
- Hydraulic Actuator

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PREMIUM
QUALITY



HIGH STRENGTH
& DURABILITY



CORROSION
RESISTANT



PRECISION
ENGINEERED

BUILT TO CONTROL. BUILT TO LAST.



TYPES OF CONTROL VALVES

NAGRAJ METAL INDUSTRIES



Control Flow • Control Pressure • Control Level • Control Temperature

1 GLOBE CONTROL VALVE



Most common type. Excellent for throttling applications. Used for precise flow control.



2 BUTTERFLY CONTROL VALVE



Compact and lightweight. Quick operation. Suitable for large flow applications.



3 BALL CONTROL VALVE



Quarter turn operation. Tight shut-off. Suitable for on/off and throttling.



4 PLUG CONTROL VALVE



Simple and reliable. Can handle dirty fluids and slurries.



5 DIAPHRAGM CONTROL VALVE



Ideal for corrosive, viscous and slurry services. No packing, leak tight.



6 PINCH CONTROL VALVE



Uses a flexible sleeve to control flow. Excellent for abrasive and corrosive media.



7 NEEDLE CONTROL VALVE



Very fine flow control. Used in low flow applications and instrumentation.



8 PRESSURE REDUCING CONTROL VALVE



Maintains downstream pressure at a set value regardless of inlet pressure changes.



9 PRESSURE RELIEF CONTROL VALVE



Protects system from excess pressure by relieving overpressure conditions.



10 3-WAY CONTROL VALVE



Mixing or diverting service. Used to control mixing of two streams or flow direction.



11 SOLENOID CONTROL VALVE



Electrically operated valve used to control the flow of fluids or gases.



12 TEMPERATURE CONTROL VALVE



Regulates temperature by controlling flow of heating or cooling medium.



13 LEVEL CONTROL VALVE



Maintains liquid level in tank or vessel by controlling the inflow of liquid.



14 FLOW CONTROL VALVE



Regulates the flow rate of liquid, gas or steam in a piping system.



15 BACK PRESSURE CONTROL VALVE



Maintains a minimum upstream pressure in the system by throttling flow.



KEY FEATURES

- ✓ Regulate process parameters accurately
- ✓ Ensure safety and system protection
- ✓ Improve efficiency and performance
- ✓ Reliable and durable construction
- ✓ Available in various designs for different applications

COMMON APPLICATIONS

- | | |
|-----------------|-----------------|
| Oil & Gas | Pharmaceutical |
| Chemical Plants | HVAC Systems |
| Power Plants | Pulp & Paper |
| Water Treatment | Food & Beverage |

COMMON ACTUATORS USED

- | | | | |
|--------------------|-------------------|--------------------|-----------------|
| Pneumatic Actuator | Electric Actuator | Hydraulic Actuator | Manual Actuator |
|--------------------|-------------------|--------------------|-----------------|



TYPES OF VALVES



NAGRAJ METAL INDUSTRIES

Control Flow • Ensure Safety • Improve Efficiency • Reliable Performance

1 GATE VALVE



Used to start or stop the flow. Best for on/off service with minimal pressure drop.



2 GLOBE VALVE



Used to regulate flow. Provides throttling action and good shut-off.



3 BALL VALVE



Uses a ball with a hole to control flow. Quarter-turn operation for quick on/off.



4 BUTTERFLY VALVE



Uses a rotating disc to control flow. Compact, lightweight and economical.



5 CHECK VALVE



Allows flow in only one direction. Automatically prevents reverse flow.



6 NEEDLE VALVE



Used for precise flow control instruments and applications.



7 PLUG VALVE



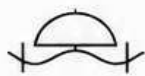
Uses a cylindrical or tapered plug with a hole to control flow.



8 DIAPHRAGM VALVE



Uses a flexible diaphragm to control flow. Ideal for corrosive or dirty fluids.



9 PINCH VALVE



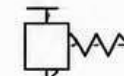
Uses a flexible sleeve to control flow. Excellent for abrasive and corrosive media.



10 SAFETY VALVE



Automatically releases pressure when it exceeds set limit to prevent system damage.



11 PRESSURE RELIEF VALVE



Releases excess pressure from a system and returns to normal when pressure drops.



12 REDUCING VALVE



Reduces high inlet pressure to lower, stable outlet pressure and maintains it.



13 SOLENOID VALVE



Electrically operated valve used to control the flow of fluids or gases.



14 ANGLE VALVE



Changes the direction of flow by 90°. Ideal for compact piping systems.



15 FOOT VALVE



Installed at the end of a suction pipe to prevent backflow and keep the pump primed.



16 3-WAY VALVE



Mixing or diverting service. Used to control mixing of two streams or flow direction.



17 4-WAY VALVE



Allows flow in four directions. Used in complex flow control systems.



18 FLOAT VALVE



Uses a float to control liquid level in tanks and reservoirs automatically.



19 LEVEL CONTROL VALVE



Maintains liquid level in a tank or vessel by controlling the inflow.



20 AIR RELEASE VALVE



Releases air accumulated in the system during operation of pipelines.



KEY FEATURES

- ✓ Regulate flow and pressure accurately
- ✓ Ensure safety and system protection
- ✓ Improve efficiency and performance
- ✓ Reliable and durable construction
- ✓ Available in various designs for different applications

COMMON APPLICATIONS

- | | |
|-----------------|-----------------|
| Oil & Gas | Pharmaceutical |
| Chemical Plants | HVAC Systems |
| Power Plants | Pulp & Paper |
| Water Treatment | Food & Beverage |

COMMON ACTUATORS USED

- | | | | |
|--------------------|-------------------|--------------------|-----------------|
| | | | |
| Pneumatic Actuator | Electric Actuator | Hydraulic Actuator | Manual Actuator |



TYPES OF STEEL SECTIONS

NAGRAJ METAL INDUSTRIES



COMMON STEEL SECTIONS USED IN CONSTRUCTION & ENGINEERING

1 I-BEAM / H-BEAM



Used in beams and load-bearing structures.

2 CHANNEL SECTION



Used in structural frameworks and supports.

3 ANGLE SECTION



Used for bracing, frames and general fabrication.

4 T SECTION



Used in partitions, frames and supports.

5 SQUARE BAR



Used in general fabrication, grills and railings.

6 ROUND BAR



Used in bolts, shafts, rod and machinery parts.

7 FLAT BAR



Used in fabrication, frames, brackets and supports.

8 HEXAGON BAR



Used in bolts, nuts, shafts and machine parts.

9 PIPE (HOLLOW ROUND)



Used in structural, plumbing, handrails and conveyance.

10 SQUARE HOLLOW SECTION (SHS)



Used in structural applications, frames and columns.

11 RECTANGULAR HOLLOW SECTION (RHS)



Used in structural, automotive and general fabrication.

12 CIRCULAR HOLLOW SECTION (CHS)



Used in structures, trusses, poles and masts.

13 Z SECTION



Used in purlins, roofing and sheeting supports.

14 C SECTION



Used in purlins, channels and structural supports.

15 RAIL SECTION



Used in railway tracks, cranes and heavy loads.

16 BULB FLAT SECTION



Used in shipbuilding, tanks and heavy structures.



NOTE: Steel sections are available in various sizes, grades and specifications to suit your requirements.



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& DURABILITY



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RESISTANT



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TYPES OF ANGLES & CHANNELS

NAGRAJ METAL INDUSTRIES



HIGH STRENGTH



PREMIUM QUALITY



CORROSION RESISTANT



DURABLE & RELIABLE

TYPES OF ANGLES

1. EQUAL ANGLE



Both legs are of equal size.

2. UNEQUAL ANGLE



Legs are of different size.

3. LONG LEG ANGLE



One leg is longer than the other.

4. SHORT LEG ANGLE



One leg is shorter than the other.

5. THICK ANGLE



Angle with thicker legs.

6. THIN ANGLE



Angle with thin legs.

7. ROUNDED EDGE ANGLE



Edges are rounded.

8. SLOTTED ANGLE



Has slots for adjustments.

9. PERFORATED ANGLE



Has holes for fixing purpose.

10. STAINLESS STEEL ANGLE



Made from stainless steel.

TYPES OF CHANNEL

1. C CHANNEL



Standard C shaped channel.

2. U CHANNEL



U shaped channel.

3. PARALLEL FLANGE CHANNEL



Flanges are parallel.

4. TAPER FLANGE CHANNEL



Flanges are tapered.

5. SLOTTED CHANNEL



Has slots for mounting.

6. PERFORATED CHANNEL



Has holes for fixing purpose.

7. STAINLESS STEEL CHANNEL



Made from stainless steel.

8. LIGHT DUTY CHANNEL



Used for light applications.

9. HEAVY DUTY CHANNEL



Used for heavy applications.

10. DEEP CHANNEL



Has more depth.

T TYPES ANGLE

1. EQUAL T ANGLE



Top and stem are of equal size.

2. UNEQUAL T ANGLE



Top and stem are of different size.

3. LONG LEG T ANGLE



Top flange is longer.

4. SHORT LEG T ANGLE



Top flange is shorter.

5. THICK T ANGLE



T angle with thick section.



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TYPES OF PERFORATED SHEET



NAGRAJ METAL INDUSTRIES

PREMIUM QUALITY STAINLESS STEEL PERFORATED SHEET, WELDMESH, WAIRMESH & ALL TYPES JALI

PERFORATED SHEET



WELDMESH (WELDED WIRE MESH)

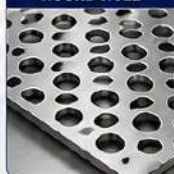


WAIRMESH (CRIMPED WIRE MESH)



ALL TYPES JALI (PERFORATED PATTERNS)

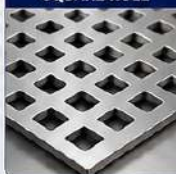
ROUND HOLE



SLOT HOLE



SQUARE HOLE



RECTANGULAR HOLE



HEXAGONAL HOLE



OBLONG HOLE



DIAMOND HOLE



TRIANGULAR HOLE



CROSS HOLE



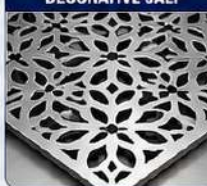
CLOVER HOLE



ROUND SLOT HOLE



DECORATIVE JALI



STAR HOLE



LOUVERED SHEET



MICRO PERFORATED



CUSTOM DESIGN



PREMIUM
QUALITY



CORROSION
RESISTANT



HIGH
STRENGTH



ACCURATE
CUTTING



SMOOTH
FINISH



WIDE RANGE OF
APPLICATIONS



READY STOCK &
CUSTOM SIZE



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TYPES OF RIVETS

NAGRAJ METAL INDUSTRIES

— STRONG. RELIABLE. PRECISION ENGINEERED. —



SOLID RIVET



Used for strong and permanent fastening.

TUBULAR RIVET



Lightweight and ideal for general fastening.

SEMI-TUBULAR RIVET



Suitable for medium-duty applications.

BLIND (POP) RIVET



Can be set from one side, ideal for blind applications.

DRIVE RIVET



Designed for high-strength assemblies.

SPLIT RIVET



Provides a firm grip and vibration resistance.

SELF-PIERCING RIVET



Pierces material and locks in place without pre-drilled holes.

FRICTION RIVET



Uses friction to create a strong and secure joint.

EXPLOSIVE RIVET



Specialized for heavy-duty and high-strength fastening.



NOTE: Rivets are used in various applications for strong, permanent and reliable fastening.



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& DURABILITY**



**CORROSION
RESISTANT**



**PRECISION
ENGINEERED**



TYPES OF RIVETS



NAGRAJ METAL INDUSTRIES

— STRONG. RELIABLE. PRECISION ENGINEERED. —

1 SOLID RIVET



Used for strong and permanent fastening.

2 SEMI-TUBULAR RIVET



Lightweight and ideal for general fastening.

3 TUBULAR RIVET



Designed for light-duty applications.

4 BLIND RIVET (POP RIVET)



Can be set from one side, ideal for blind applications.

5 DRIVE RIVET



Designed for high-strength assemblies.

6 SPLIT RIVET



Provides a firm grip and vibration resistance.

7 EXPANSION RIVET



Expands during setting for strong holding.

8 STRUCTURAL RIVET



Used in high-stress structural joints.

9 BIFURCATED RIVET



Legs split to provide strong anchoring.

10 TRIFURCATED RIVET



Three-way expansion for extra holding power.

11 MULTI-GRIP RIVET



Accommodates a wide grip range.

12 PEEL RIVET



Ideal for soft or brittle materials.

13 COUNTERSUNK RIVET



Sits flush with surface for a smooth finish.

14 FLAT HEAD RIVET



Provides a larger bearing surface.

15 ROUND HEAD RIVET



General purpose with a rounded head.

16 BUTTON HEAD RIVET



Low-profile and aesthetic finish.

17 PAN HEAD RIVET



Wide head for better load distribution.

18 DOME HEAD RIVET



Decorative head with strong fastening.

19 CHERRY RIVET



Commonly used in aircraft and sheet metal.

20 BRAZIER RIVET



Used in plumbing and metalworking.

21 MAG RIVET



High strength and vibration resistant.

22 WELD RIVET



Used for welding applications.

23 FRANGIBLE RIVET



Breaks off after setting for tamper protection.

24 ROLL RIVET



Designed for heavy-duty and permanent joints.

 **NOTE:** Rivets are used in various applications for strong, permanent and reliable fastening.



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PREMIUM QUALITY



HIGH STRENGTH & DURABILITY



CORROSION RESISTANT



PRECISION ENGINEERED



TROLLEY WHEEL TYPES



NAGRAJ METAL INDUSTRIES

⚙️ PREMIUM QUALITY • HIGH DURABILITY • SMOOTH PERFORMANCE

1 CAST IRON WHEEL



Used in heavy-duty industrial applications.

2 STEEL WHEEL



High strength and durability for heavy loads.

3 POLYURETHANE WHEEL



Excellent shock absorption and floor protection.

4 NYLON WHEEL



Lightweight and corrosion resistant.

5 RUBBER WHEEL



Provides grip and absorbs vibrations.

6 PHENOLIC WHEEL



Heat resistant and suitable for high loads.

7 V-GROOVE WHEEL



For V-groove tracks and alignment.

8 U-GROOVE WHEEL



Used in U-channel tracks and rails.

9 FLAT WHEEL



For general purpose applications.

10 DOUBLE FLANGE WHEEL



Extra stability on parallel tracks.

11 SINGLE FLANGE WHEEL



For one-side track guidance.

12 ADJUSTABLE FLANGE WHEEL



Flange can be adjusted as per track.

13 CONICAL WHEEL



For tapered tracks and conical rails.

14 BALL BEARING WHEEL



Smooth and efficient rolling performance.

15 HEAVY DUTY WHEEL



Built for extreme loads and tough conditions.

16 HIGH TEMPERATURE WHEEL



Suitable for high heat industrial environments.

17 GUIDE ROLLER WHEEL



Used to guide and support movement.

18 CHAIN GUIDE WHEEL



For chain tracks and guide systems.

19 TRACK WHEEL



Used in cranes and heavy machinery tracks.

20 POLYAMIDE WHEEL



Strong, lightweight and chemical resistant.

⚙️ **NOTE:** Trolley wheels are used in material handling systems, trolleys, conveyors, cranes, gates, and industrial applications for smooth and reliable movement.



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PREMIUM
QUALITY



HIGH STRENGTH
& DURABILITY



CORROSION
RESISTANT



PRECISION
ENGINEERED



TYPES OF PULLEYS



— USED TO CHANGE DIRECTION OF FORCE AND TRANSMIT POWER —



SMOOTH OPERATION



HIGH EFFICIENCY



DURABLE
CONSTRUCTION



CORROSION
RESISTANT



LONG LASTING
PERFORMANCE

1. FIXED PULLEY



The pulley is fixed in one place and only changes the direction of the force.

2. MOVABLE PULLEY



The pulley moves with the load and reduces the effort required to lift.

3. SINGLE GROOVE PULLEY



Has one groove for a single belt. Used in light to medium applications.

4. DOUBLE GROOVE PULLEY



Has two grooves for two belts. Provides better grip and more power.

5. MULTI GROOVE PULLEY



Has multiple grooves for multiple belts. Used for high power transmission.

6. STEP PULLEY



Consists of multiple different diameter steps for variable belt speed.

7. TIMING PULLEY



Used with timing belts to maintain synchronization without slippage.

8. V-BELT PULLEY



Has a V-shaped groove to fit V-belts. Provides high grip and power.

9. FLAT BELT PULLEY



Used with flat belts for power transmission over long distances.

10. IDLER PULLEY



Used to guide and maintain belt tension. Does not transmit power.

11. GUIDE PULLEY



Guides the belt in the correct path and prevents lateral movement.

12. CONICAL PULLEY



Used with flat belts in conical bushes. Allows easy speed adjustment.

13. ROPE PULLEY



Designed with a groove suitable for wire ropes. Used in lifting and hoisting applications.

14. CROWN PULLEY



Has a crowned surface to keep the belt centered and prevent misalignment.

15. COMPOUND PULLEY



Combination of two or more pulleys in one unit for different speed ratios.



NOTE: Pulleys help transmit power efficiently, change direction of force, and increase mechanical advantage in various applications.



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TYPES OF PIPE NIPPLE

NAGRAJ METAL INDUSTRIES



Pipe nipple is a short length of pipe with male threads on one or both ends used to connect two fittings.

1. CLOSE NIPPLE



- Threads on both ends.
- Used for connecting two female threaded fittings.
- Available in various lengths.

2. HEX NIPPLE



- Hexagon in the middle for wrench grip.
- Easy to tighten and remove.
- Used in tight spaces.

3. REDUCING NIPPLE



- Different thread sizes on each end.
- Used to connect fittings of different sizes.
- Available in various combinations.

4. LONG NIPPLE



- Longer length than close nipple.
- Used when more distance is required between fittings.
- Available in various lengths.

5. HALF NIPPLE



- Threaded on one end only.
- Used to connect to a female threaded fitting on one side.
- Widely used in plumbing and piping.

6. REDUCING HEX NIPPLE



- Hex in the middle with different thread sizes.
- Provides wrench grip and size reduction in one piece.

7. BARREL NIPPLE



- Longer body with threads on both ends.
- Used in high pressure and long line applications.
- Strong and durable construction.

8. SWAGE NIPPLE



- One end swaged (larger) and the other end regular size.
- Used to connect pipe to valves, pumps and fittings.

9. BULL NIPPLE



- Hex (bull nose) in the middle.
- Provides strength and wrench grip.
- Used in heavy duty applications.

10. STREET NIPPLE



- Threaded on one end, plain on other.
- Screwed into a female fitting and welded on the other end.
- Common in branch connections.

11. EXTRA LONG NIPPLE



- Extra long length for special applications.
- Used in tanks, boilers and long piping systems.

12. WELD NIPPLE



- Plain on both ends.
- Used for welding in piping systems.
- Provides leak proof connection.

13. HEX REDUCING NIPPLE



- Hex body with reducing thread sizes.
- Used where size reduction and wrench grip are required.

14. PIPE x NPT NIPPLE



- One end pipe thread (BSPP), other end NPT thread.
- Used to connect different thread types.

15. DOUBLE END NIPPLE



- Threads on both ends with extra length.
- Used to connect two fittings with more spacing.



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TYPES OF SLEEVES



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1. PLAIN SLEEVE



2. FLANGED SLEEVE



3. SHOULDER SLEEVE



4. TAPER SLEEVE



5. SPLIT SLEEVE



6. THREADED SLEEVE



7. KEYED SLEEVE



8. DOWEL SLEEVE



9. EXPANSION SLEEVE



10. DISTANCE SLEEVE



11. REDUCTION SLEEVE



12. LOCK SLEEVE



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TYPES OF BUSHES



NAGRAJ METAL INDUSTRIES

1. PLAIN BUSH



Simple, cost-effective bush. Provides sliding support and reduces friction.

2. FLANGED BUSH



Has a flange to locate the bush axially and prevent movement.

3. THRUST WASHER BUSH



Designed to take axial (thrust) loads in one direction.

4. SPLIT BUSH



Slit design for easy installation and removal without dismantling.

5. SELF-LUBRICATING BUSH



Contains solid lubricants. Works well with minimal or no external lubrication.

6. OIL-IMPREGNATED BUSH



Porous structure holds oil. Provides long life and smooth operation.

7. RUBBER BUSH



Reduces vibration, noise and shock. Used in automotive suspension and mounts.

8. POLYURETHANE BUSH



High wear resistance and load capacity. Used in high stress conditions.

9. NEEDLE ROLLER BUSH



Uses needle rollers for high load carrying capacity in limited space.

10. LINEAR MOTION BUSH



Used in linear motion applications. Ensures smooth and accurate movement.

11. TAPER BUSH



Tapered design for locking onto a shaft. Provides strong grip and easy mounting.

12. QD BUSH



Quick Disconnect bush. Allows easy mounting and removal of components.

13. GUIDE BUSH



Guides moving parts with high accuracy. Used in dies, moulds and punches.

14. COMPOSITE BUSH



Combination of metal and polymer for low friction and high durability.

15. PTFE BUSH



Chemically resistant and low friction. Used in corrosive and high temperature applications.

16. SPHERICAL BUSH



Allows angular movement and misalignment. Used in oscillating and pivot joints.



NOTE: Bushes are used to reduce friction, support loads, guide motion, absorb vibration and improve performance and life of mechanical systems.

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TYPES OF CHAINS



NAGRAJ METAL INDUSTRIES

1 ROLLER CHAIN



- Most common type
- High load carrying capacity
- Used in power transmission

2 BUSH CHAIN



- Simple & cost effective
- Used in conveyor systems
- Low to moderate load

3 SILENT CHAIN



- Smooth & quiet operation
- High speed applications
- Automotive & machinery

4 LEAF CHAIN



- Built with leaf plates
- High tensile strength
- Used in heavy machinery

5 PEG CHAIN



- Pegs attached for extra attachments
- Used in conveying systems
- Agricultural & industrial use

6 CONVEYOR CHAIN



- Attachments for conveying
- Handles heavy loads
- Material handling systems

7 HOLLOW PIN CHAIN



- Pins are hollow for attachments
- Used in agricultural implements
- Easy to customize

8 EXTENDED PIN CHAIN



- Pins extended on one or both sides
- Used for special attachments
- Conveyor & lifting applications

9 DROP FORGED CHAIN



- Extremely strong & durable
- Withstands high stress
- Used in harsh environments

10 ENGINEERING STEEL CHAIN



- Made from alloy steel
- High strength & wear resistance
- Industrial & mechanical use

11 STAINLESS STEEL CHAIN



- Corrosion & rust resistant
- Used in wet & corrosive areas
- Food, marine & chemical use

12 PLASTIC CHAIN



- Lightweight & corrosion proof
- Low friction & noise
- Food processing & packaging

13 DOUBLE PITCH CHAIN



- Twice the pitch of roller chain
- Used in conveyors & lifts
- Handles heavy loads

14 TRIPLE PITCH CHAIN



- Three times the pitch
- Used in heavy duty conveyors
- Long distance conveying

15 SIDE BOW CHAIN



- Bow shaped side plates
- Smooth movement
- Material handling systems

16 TOP CHAIN



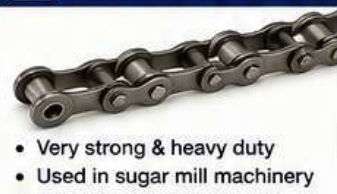
- Flat top for stability
- Used in bottle conveying
- Packaging industries

17 SLAT CHAIN



- Slats for carrying load
- Used in heavy conveyors
- Mining & bulk handling

18 SUGAR MILL CHAIN



- Very strong & heavy duty
- Used in sugar mill machinery
- High shock load capacity

19 DRAG CHAIN



- Designed for dragging
- Used in lifting & dragging
- Industrial applications

20 HOOK CHAIN



- Hooks for lifting & slinging
- Used in material handling
- Lifting & pulling applications



NOTE: Select the right chain based on load, speed, environment and application to ensure performance, safety and long service life.



TYPES OF GASKETS



NAGRAJ METAL INDUSTRIES

1. FLAT GASKET



Simple and most common gasket. Used between flat surfaces in pipes, valves, pumps and compressors.

2. RING GASKET



Used in high pressure and high temperature applications. Common in boilers and heat exchangers.

3. SPIRAL WOUND GASKET



Made of metal strip and filler material wound together. Suitable for high pressure, temperature and corrosive applications.

4. KAMMPROFILE GASKET



Metal core with soft sealing facing. Used in heat exchangers, reactors and high pressure systems.

5. CORRUGATED METAL GASKET



Corrugated shape provides better sealing at high temperatures and pressures.

6. RUBBER GASKET



Made of rubber materials. Used in low pressure applications for water, air and general services.

7. FIBER GASKET



Made of compressed fibre materials. Used in water, oil, gas and steam applications.

8. PTFE (TEFLON) GASKET



Made of PTFE material. Excellent chemical resistance and used in corrosive applications.

9. GRAPHITE GASKET



High temperature resistant and chemically inert. Used in steam, oils, acids and chemical services.

10. METALLIC GASKET



Made of soft metals like copper, aluminium, etc. Used in high temperature and pressure applications.

11. COMPOSITE GASKET



Made of combination of materials for better sealing. Used in wide range of industrial applications.

12. ENVELOPE GASKET



Metal outer with soft filler material. Used in flange and heat exchanger applications.

13. LENS GASKET



Lens shape increases sealing ability. Used in high pressure and high temperature applications.

14. JACKETED GASKET



Filler material jacketed with metal. Used in high pressure and corrosive environments.

15. RTJ (RING TYPE JOINT) GASKET



Used in very high pressure and high temperature services like oil & gas and petrochemical industries.

16. LIQUID GASKET / SEALANT



Semi-liquid material applied between surfaces. Used for low pressure and irregular surface joints.

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TYPES OF INDUSTRIAL HINGES



NAGRAJ METAL INDUSTRIES

1. BUTT HINGE



Commonly used for doors, cabinets and general purpose applications.

2. CONCEALED HINGE



Hidden from view when door is closed. Used in cabinets and furniture.

3. PIANO HINGE (CONTINUOUS HINGE)



Runs the full length of the door or lid. Provides uniform strength and support.

4. WELD-ON HINGE



Welded directly to metal frames or gates. Strong and suitable for heavy use.

5. DETACHABLE HINGE



Pin can be removed to detach the door or panel easily.

6. SPRING HINGE



Automatically closes the door after opening. Used in doors that need self-closing.

7. TORQUE HINGE



Provides controlled movement and holds position. Used in electrical panels and lids.

8. HEAVY DUTY HINGE



Designed for heavy doors, gates and industrial applications.

9. LIFT-OFF HINGE



Door can be lifted off by removing the pin. Used in heavy duty applications.

10. STRAP HINGE (TEE HINGE)



Long strap design for extra strength. Used in gates, sheds and heavy doors.

11. GATE HINGE



Adjustable hinge used in gates and doors that may sag over time.

12. FRICTION HINGE



Provides adjustable resistance for smooth movement. Used in covers and flaps.

13. PIVOT HINGE



Allows doors to pivot from top and bottom. Used in heavy doors.

14. BALL BEARING HINGE



Smooth operation with ball bearings. Used in doors requiring high performance.

15. DOUBLE ACTION HINGE



Allows door to open in both directions. Used in saloon doors and swing doors.

16. GLASS DOOR HINGE



Designed for glass doors. Provides secure and sleek installation.

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FASTENER TO DRILL BIT



MATCH DRILL BIT SIZE TO FASTENER

FASTENER (BOLT / SCREW)

DRILL BIT SIZE

M6



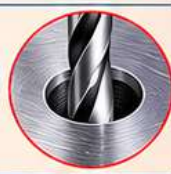
10 mm



M8



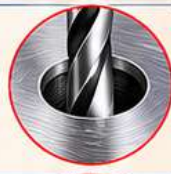
12 mm



M10



14 mm



M12



18 mm



M16



22 mm



M20



28 mm



M24



34 mm



MATCH DRILL BIT SIZE TO FASTENER FOR ACCURATE INSTALLATION



ACCURATE FIT
Perfect Hole Size



STRONG HOLD
Better Tightening



PRECISE DRILLING
Clean & Accurate



PROFESSIONAL
Quality Results

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TYPES OF PIPE THREADS



NAGRAJ METAL INDUSTRIES

1 **NPT**
(NATIONAL PIPE THREAD)



2 **NPTF**
(DRYSEAL PIPE THREAD)



3 **BSPP**
(BRITISH STANDARD
PIPE PARALLEL)



4 **BSPT**
(BRITISH STANDARD
PIPE TAPER)



5 **METRIC**
PIPE THREAD



6 **UN/UNF**
THREAD



7 **PG**
THREAD



8 **G**
THREAD



9 **R**
THREAD



10 **Rc**
THREAD



11 **Rp**
THREAD



12 **NPS**
(NATIONAL PIPE
STRAIGHT)



13 **NPSC**
(NATIONAL PIPE
STRAIGHT COUPLING)



14 **NPSM**
(NATIONAL PIPE
STRAIGHT MECHANICAL)



15 **ACME**
THREAD



16 **BUTTRESS**
THREAD



KNOW THE THREAD, MAKE IT STRONG.
RIGHT THREAD – RIGHT CONNECTION – ZERO LEAKAGE.



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**PREMIUM
QUALITY**



**HIGH STRENGTH
& DURABILITY**



**CORROSION
RESISTANT**



**PRECISION
ENGINEERED**